

John B. Goodenough Biography

Introduction

The Nobel laureate (Nobel Prize Winner) John B. Goodenough along with other scientist, invented the revolutionary Lithium-ion Battery. It was key effect in evolving the world's energy resources, as the battery contains the ability to regain charge, therefore making the lithium-ion battery sustainable and reusable. This battery paved the way to bring out the devices that we utilise today, mainly phones and laptops. The invention and the Nobel prize were merely the last stepping stone of Goodenough's achievements as he has helped paved the future that we have today. This biography will delve deep into his life and remarkable research of John B. Goodenough.



Figure 1. Photo of John B. Goodenough. 2023.

Upbringing

Born in Germany, from American parents on July 25th, 1922, John lived in America. John comes from an academic family as John's father was an Oxford University graduate (Wikipedia contributors, 2024). John would take upon this academic trait as well, becoming very successful in the future as a scientist. However, from young John suffered from dyslexia, a condition making it near impossible to read or write. Dyslexia at the time was a mystery to the medical arsenal in the 1920s having no treatment for John. Therefore, John had a difficult time throughout his duration at the Groton School. He was considered a "Backwards



Figure 2. photo of John B. Goodenough as he was drafted for WW2.

Student" although John B Goodenough proved people wrong, teaching himself simple steps such as learning to write (Wikipedia contributors, 2024). Not long after John had risen to the top of his class, overcoming the odds. After his time in Groton, he transferred around a few different schools. Then graduated from Yale receiving his degree in Mathematics from there.

Near the time he graduated from Yale, 1943, John B. Goodenough was enrolled to help the USA during WW2 with the position of meteorologist. He served his time until 1945 for 2 years. Once WW2 ended John returned to his

education. Entering the University of Chicago. During his time in

the University of Chicago he met a Canadian history graduate named Irene Wiseman. They got married a year before John Goodenough would obtain his Ph.D. Throughout his 5 years in the University of Chicago John had gained a master's and a Ph.D. in physics, the year being 1952. John B (Wikipedia contributors, 2024). Goodenough was around 30 years old by the time he graduated and John's Research and inventions are soon to come after this second graduation.

John's Career

After his second graduation, he came to MIT Lincoln Laboratory, for 24 years. On John B. Goodenough's first massive project he worked and lead a team for research and development of MRAM or also known as Magnetoresistive Random Access Memory. Different from RAM this type of memory access is non-volatile meaning a computer can store data even when turned off, which is a vital point in technology advancement. Allowing us in BISB to note take and use electronics at greater lengths for learning. Other than being the leader for the research team, John was working more specifically on magnetism and on the metal-insulator transition behaviour in transition-metal oxides (Wikipedia contributors, 2024). A metal-insulator transition is the process and change of a metal into an insulator, therefore John would explore how turning metal oxides into insulators would differ from normal metal-insulator transition. His effects on the research and development on MRAM were notable however this research would lead John B. Goodenough to eventually develop the well-known **Goodenough-Kanamori Rules**.

Referring to the previous paragraph, John B. Goodenough soon after the research of MRAM would develop the Goodenough-Kanamori Rules. The development of these rules all began during his duration at the MIT Lincoln Laboratory, as the rules do relate to his research of the magnetism. The major rules that are including in the Goodenough-Kanamori rules is based of 4 basis pillars. The 4 bases can be put into simplified words:

The 4 Pillar Bases of the complex Goodenough-Kanamori rules explain and explore how materials magnetic behaviour acts influenced by the electron interactions helping to determine if the material is magnetic or not (Goodenough, 2008). After 24 years in MIT Lincoln, the year 1976, John moved to a university for research. The University of Oxford. The reason that John B. Goodenough had moved was mainly due to funding being cut off



Figure 3. Image of the MIT Lincoln Laboratory during the year 1952, when John B. Goodenough was conducting his research.

from the government. I shall explore the events that will occur during his time in the University of Oxford.

Stated before, during the late 1970s John had moved to the Oxford University for more research. He would say for a decade before continuing on. John B. Goodenough quickly became the Head of the Inorganic Chemistry Department (*Professor John Goodenough Wins*



Figure 4. Image of the Inorganic Chemistry Department, present day.

2019 Nobel Prize in Chemistry For, 2019). He along with other began research and the development of the iconic **Rechargeable Lithium-ion battery**. The battery was a major invention along with the Goodenough-Kanamori Rules. For the Battery had the special property to recharge itself. The recharging occurs from the lithium-ions being transport throughout the battery. Before this batteries were a one and done type of

power for devices and running machines. So, the invention of the battery allowed people to save money and time with their rechargeability. The battery also possessed other features; however, these aren't of notice. The breakthrough like John B. Goodenough's older products was not a single man consist of many people collaborating. These were Koichi Mizushima, Philip C Jones and Philip J Wiseman (*Professor John Goodenough Wins 2019 Nobel Prize in Chemistry For, 2019*). Moreover, not long after the Lithium-ion battery, 1980, the battery was improved by expanding its capacity by double the amount. Other than the Kanamori-Goodenough Rules, this were to be his finest invention throughout his years of research. Unfortunately, the inventions took time to begin its globalisation because of the denied patent (legal documentation, to prevent others from manufacturing or selling the invention) from Oxford due to insufficient funds from paying his academic salary.

Therefore, John went looking for someone to pay for his patent. John struggled to find someone to write up this patent because he would have zero royalty on his invention. Although Goodenough had no other options and reluctantly agreed. Until the license was given to Sony Corporations as Sony did promise to decide to pay royalties to John and the other inventors (Wikipedia contributors, 2024). A person working for Sony, Akira Yoshino improved on the design for both efficiency and easier manufacturing. Where after John would receive the Japan Prize, a prize for technological or science (The reason John received the award was because Sony is a Japanese company) and the Nobel Prize. However, these appointment of the awards were not until later. 2011 and 2019 respectively.

Now in the year 1986, once again John B. Goodenough moved to Texas university. During his time in this university, John was very innovative. Contributing to the topic sustainability and electrical. While John was at Texas University, he proceeded to aid in electrical engineering, advancing electronics specifically. There is a quote from the president of the Texas



Figure 5. Image of Texas University, 1986.

University, “Billions of people in the world benefit every day from John’s innovations.” proving how relevant he is in the past and present.

Now during 1986, John B. Goodenough had reached the age of 64. However, John still attempted to push the boundaries as he stayed in

Texas University. slowly but surely expanding the world of batteries, specifically refining rechargeable batteries as well as inventing new different types of batteries. Even if John had no outstanding achievements like his invention of the lithium-ion battery, it is important not to disregard that John was still active and continued to contribute towards more electrical advancements.

John's time in Texas University is greater than any time in his studies, for Goodenough would continue researching in this university for 36 years. More specifically, Goodenough didn't leave Texas University before he died, dying at the age of 100. In 2016, John B Goodenough's wife unfortunately passed away at the age of 70 because of a development dementia.

However, a year after the death of Goodenough's wife (2017), John expanded battery technology, for example the glass battery. the glass battery is when a type is glass made by combining lithium-hydroxide and lithium-chloride as well as a little bit of barium. Therefore, making a battery that has similar qualities of the rechargeable battery, although not producing harmful metal formation. Leading to the Glass battery being more efficient than other types (*Glass Battery - a Peculiar Development in Our Understanding of Physics or the Key to the Next Energy Revolution?*, n.d.). This being one of John's last inventions.

Not long after the events of the invention of the glass battery; previously mentioned above John B Goodenough was due to get the **Nobel Prize** and to become a Nobel laureate. John and his other colleagues were awarded the Nobel Prize in 2019.

Conclusion



Figure 6. Picture of John B Goodenough at the age of 100. Before Passing away.

On June 25, 2023; At age of 100 and merely one month until his 101st birthday. John B Goodenough passed away. Goodenough was a monumental character in the ever-changing state of electricity and batteries. Fighting for the USA in World War 2, working in collages ever since the age of early 30s to 100 years old. However, John's work and passion will never be forgotten for the numerous invention made and assisted paved the way for the Present but as well as the near future. BISB couldn't possibly ignore this man's research, for it has aided all of us by using electronics of all sorts.

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